

Group Name: Akagera

Project Topic: Build a Small Language Model for Agriculture and Climate Change in Nigeria

Problem Statement

Agriculture remains the backbone of Nigeria's economy, providing livelihoods for over 70% of the population and contributing significantly to food security. However, the sector faces mounting challenges from climate change, including irregular rainfall, flooding, soil degradation, and rising temperatures. These environmental shifts threaten crop yields, livestock productivity, and the resilience of rural communities. Farmers, policymakers, and researchers urgently need localized, accessible, and actionable knowledge to adapt practices and mitigate risks.

Despite the abundance of global research on climate change and agriculture, most existing language models are trained on data that is not contextually relevant to Nigeria's unique ecological zones, farming practices, and socio-economic realities. This creates a knowledge gap: farmers and stakeholders often lack tailored insights in local languages and culturally appropriate formats. Furthermore, limited access to digital tools in rural areas exacerbates the challenge of disseminating climate-smart agricultural information.

Therefore, there is a pressing need to develop a small, domain-specific language model focused on agriculture and climate change in Nigeria. Such a model would synthesize localized data, provide context-aware recommendations, and support decision-making for farmers, extension workers, and policymakers. By bridging the gap between global climate knowledge and local agricultural realities, this tool can empower communities to build resilience, improve productivity, and safeguard food security in the face of climate uncertainty.